Q.1 Types of operating systems.

1] Batch operating system.

Defination:
Executes Jobs in batches without user interaction.

uses: Payroll, billing

advantage: No user intereaction

~~slower~~

disadvantage: slow response.

2] Time sharing operating system

Defination: Allows many users to share one computer.

use: universities, office

advantage:- fast responsive, support many user

disadvantage: security issue, system may slow down.

3] multiprograming operating system

defination: Keeps mutiple programs in memory to use the CPU efficiertly

uses: Business systems
advantage: Better CPU Utilization
Disadvantage: complex to manage.

4] multitasking operating system

defination: Runs multiple applications at the same time

uses: personal computer

advantages: save time, improves productivity

disadvantages: need more RAM & CPU

5] multiprocessing operating system :-
defination: uses two or more processors together

uses:- servers, scientific computing
advantage:- faster processing

disadvantage: expensive hard more.

6) Realtime operating system (RTOS)

defination: responds to tasks within a fixed time

uses: Robots, medical services

advantage: fast & accurate

disadvantage: costly & complex

7] Distributed operating system

defination: connects many computer & works as one system

uses - cloud computing

advantage: resource sharing

8) network operating system

defination: manage computer connected through a network uses! schools, office

advantage: easy file & printer sharing

disadvantage: network failure office.

g) mobile operating system

defination: os for smartphone & tablets

uses: mobile device

advantage: ~~~~ easy to use

disadvantage: limited compared to desktop os.

Q. 2 Types of system calls.

1] Processor control:-
~~to~~ creats, executes & manage process.

2] file managment
creates, open, reads, writes with hardware device

3] memory managment
Allocates ~~of~~ & frees memory

4] Information managment:-
Gets or sets system & process information

5] communication:
Allows process to exchange

7] protection & security
controls permission & access to resources.

Q.3 System call definations

process control

fork () → create a new child process

vfork () → create a child process that share the parents memory until exec() or exit.

clone() → create new process or thread

execve() → Replaces the current process with a new program

exit () → ends a process immediate

wait () → wait for any child process

waitpid() → waits for child process & return detailed status.

kill() → sends a signal to process.

getpid() → ~~reet~~ return a current process ID

getppid() return the parents process ID

nice () → changes a process priority

setpriority () → sets process priority

getpriority () → gets process priority

file managment :

open () → open a file
create () → create a file
close () → close a file
read () → reads data from file
write () → writes data to a file
pread () → reads from specific file position
pwrites () → writes to a specific file position

lseek () → changes the file pointer position
slat () → gets file info
fslat () → gets into about a symboli
access () → check files access permission
unlink () → rename a file or directory

mkdir () → creates a directory

rmidir() → remove an empty directory

dup() → Duplicates a file descriptor

dup2() → duplicates a file describe to specified number

fcntl() → controls file descriptor operations

fsync() → Save file data & metadata to disk

fdatasync() → save only file data to disk

truncate() → changes the size of file

ftruncate() → changes the size of an open file.